

$$L = \lim_{n \rightarrow \infty} \frac{2^{n^2}}{n^{2n}}$$

$$\lim_{n \rightarrow \infty} \ln \left(\frac{2^{n^2}}{n^{2n}} \right) = \lim_{n \rightarrow \infty} \left(\ln(2^{n^2}) - \ln(n^{2n}) \right)$$

$$\lim_{n \rightarrow \infty} \left(n^2 \ln(2) - 2n \ln(n) \right)$$

$$\lim_{n \rightarrow \infty} (n^2 \ln(2)) - \lim_{n \rightarrow \infty} (2n \ln(n))$$

$$n^2 \ln(2) \Rightarrow \text{complejidad } n^2$$

$$2n \ln(n) \Rightarrow \text{complejidad } n \log(n)$$

$$n^2 \gg n \log(n)$$

$$\Rightarrow L = \lim_{n \rightarrow \infty} \frac{2^{n^2}}{n^{2n}} \Rightarrow \infty$$
